

SEMICON INDIA HACKATHON 2026

# Drift-Sense

AI-Powered Navigation-Error Recovery for Wafer Inspection Tools

**Team Blitzkrieg**

Bhushan Thatte & Yash Kothari

BITS Pilani — Electronics & Electrical Engineering

Applied Materials — Problem Statement 2

# Navigation-Error Recovery, Not Template Matching

Given a **high-resolution reference** (1000×1000, 1 nm/px, 100× zoom) and a **wide-search image** (1000×1000, ~10 nm/px, 10× zoom), locate the reference pattern's center to **sub-pixel precision**.

## Key Reframing

The tool *knows* where it commanded the stage. Drift is a **small, bounded error** — not an arbitrary location in a 10  $\mu\text{m}$  field. A *recovery* system exploits this prior, not just image content alone.

## Phase 1 Constraints

- ▶ Exact 10× zoom ratio (known)
- ▶ No rotation
- ▶ Reference always present

## Phase 2 Extensions

- ▶ Zoom unknown — uniform [8×, 12×]
- ▶ Rotation unknown —  $\pm 5^\circ$
- ▶ ~20% absent pairs (rejection required)
- ▶ Degraded images (charging, defocus, noise)
- ▶ 20s hard timeout,  $\leq 5\text{s}$  median target

# Physically-Motivated SEM Image Synthesis

## Structure Rendering

- ▶ **DRAM** — word-line / bit-line / contact array
- ▶ **FinFET** — parallel fin / gate-line array
- ▶ **RGB Optical** — 3-channel diffraction + chromatic aberration

Same renderer at both scales — only pitch/width parameters change with zoom ratio.

## Line-Edge Roughness (LER)

Smoothed random perturbation of each edge — not pixel-IID noise but spatially correlated, matching real lithographic processes. Makes neighbouring "identical" cells visually similar but *not pixel-identical* — the source of genuine periodic ambiguity.

## SEM Noise Model (4 Components)

- ▶ **SHOT** Poisson noise — signal-dependent, dominant
- ▶ **READ** Gaussian — additive, signal-independent
- ▶ **BLUR** Beam-spot PSF — Gaussian convolution
- ▶ **JITTER** Per-row scan-line intensity offset

Search image deliberately noisier (lower gain, larger blur) — modeling fast survey scan vs. careful characterization.

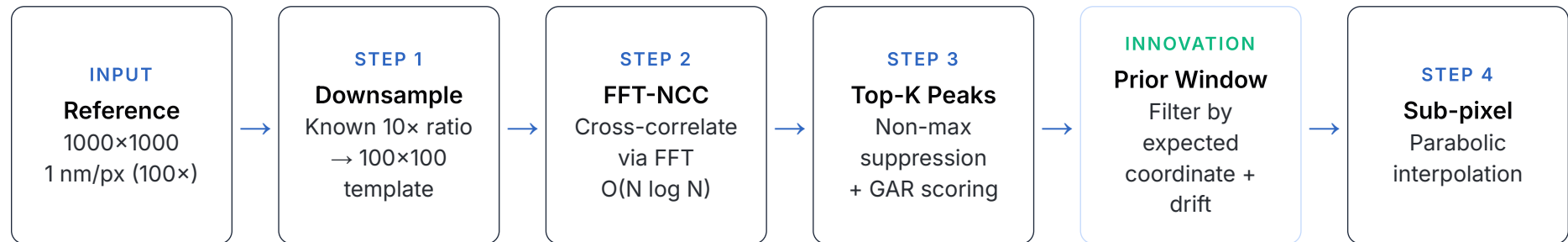
## Phase 2 Degradation Effects

- ▶ Charging artifacts (localized bright blooming)
- ▶ Scan distortion (smooth 2D displacement field)
- ▶ Polygon CD jitter ( $\pm 20\%$  critical dimension)
- ▶ Elevated shot noise + defocus

# Every Design Choice Justified Against Public Sources

| DESIGN CHOICE                 | SOURCE(S)                                                                                                         | CODE LOCATION                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Poisson-Gaussian SEM noise    | Avci et al. (shot-noise estimation); ScienceDirect M-Denoiser 2023; Oxford Academic SEM noise classification 2025 | <code>apply_sem_noise()</code>                          |
| Line-Edge Roughness (LER/LWR) | Bunday/Bishop/Villarrubia (NIST/SEMATECH); ITRS targets (<1.7 nm sub-20 nm nodes)                                 | <code>_smooth_noise_1d()</code>                         |
| FinFET pitch/width            | IEEE "Scaling of SOI FinFETs" — fin pitch 40 nm, gate length 20 nm                                                | <code>render_finfet_pattern()</code>                    |
| Optical chromatic aberration  | Nikon MicroscopyU (diffraction barrier); Wadsworth Center microscopy glossary                                     | <code>to_rgb_optical()</code>                           |
| Fast NCC via FFT              | Lewis, "Fast Normalized Cross-Correlation," Vision Interface 1995                                                 | <code>ncc_map()</code> / <code>cv2.matchTemplate</code> |
| Peak-to-Sidelobe Ratio (PSR)  | Bolme et al. MOSSE tracker, CVPR 2010                                                                             | <code>peak_to_sidelobe_ratio()</code>                   |
| Prior-window restriction      | US Patent 7,545,497; Coarse-to-fine tracking (arXiv 2012.12784); Die-bonding patent 12,327,739                    | <code>expected_xy</code> / <code>max_drift_px</code>    |

# Classical FFT-NCC Matched Filtering



## Prior-Window Innovation

The tool knows the commanded stage position. Drift is bounded (typically <60 px). Discarding candidates outside this window is simultaneously:

- ▶ **Faster** — smaller effective search area
- ▶ **More robust** — distant periodic decoys eliminated
- ▶ **Zero precision loss** — same FFT-NCC + refinement inside window

Grounded in US Patent 7,545,497 — alignment routines for optically based tools.

## Global Ambiguity Ratio (GAR)

A corrected confidence metric replacing local PSR. Compares best NCC peak vs. best *non-local* competing peak anywhere in the image (Lowe's ratio test style).

- ▶ GAR ≈ 0.88–0.91 → easy (clear unique match)
- ▶ GAR ≈ 0.98–0.999 → ambiguous (decoy present)

PSR only sees local sidelobes — wrong scope for distant periodic decoys.

# 100% Accuracy with Sub-Pixel Precision

| CONFIGURATION                    | SUCCESS RATE  | MEAN ERROR | MEDIAN ERROR | COMPUTE TIME |
|----------------------------------|---------------|------------|--------------|--------------|
| Non-adversarial (easy)           | 100% (30/30)  | 0.03 px    | 0.03 px      | ~309 ms      |
| Adversarial decoy, no prior      | 46.7% (14/30) | 194.9 px   | 155.9 px     | ~224 ms      |
| Adversarial decoy + prior window | 100% (30/30)  | 0.02 px    | 0.02 px      | ~215 ms      |

## DL Ablation — Siamese CNN (69K params)

| CONFIGURATION           | SUCCESS | MEAN ERROR | TIME     |
|-------------------------|---------|------------|----------|
| Raw DL argmax           | 6.7%    | 73.6 px    | 4.4 ms   |
| DL + NCC refinement     | 50.0%   | 66.1 px    | 7.1 ms   |
| DL + refinement + prior | 86–90%  | ~5–8 px    | 10–49 ms |

DL is 4–20× faster but hasn't converged to match classical accuracy within available compute budget. Classical FFT-NCC submitted as primary method.

## Honest Failure Case (Seed 24)

ERROR: 753 PX    WITHOUT PRIOR

True site scored **0.8464** vs decoy's **0.8438** — a 0.3% gap. Algorithm chose the decoy (closer to center, per spec tie-break rule).

**Root cause:** Purely image-based tie-break cannot resolve structurally identical periodic sites. This is the *exact failure mode* the prior-window mechanism resolves.

With prior window → 0.02 px error ✓

# Multi-Stage Beam Search over (Scale, $\theta$ , Position)

## Stage 1 — Coarse Grid Sweep

**9 scales**  $\times$  **5  $\theta$  anchors** = 45 evaluations

Scale: {8.0, 8.5, 9.0, ..., 12.0}  $\cdot$   $\theta$ : {-5, -2.5, 0, 2.5, 5} $^\circ$

Each eval: `cv2.matchTemplate(TM_CC0EFF_NORMED)`

Output: **Top-3 (scale,  $\theta$ )** by NCC score

## Stage 2 — $\theta$ Refinement

Per beam candidate: sweep  $\theta \pm 1.5^\circ$  in **0.5 $^\circ$  steps**

Output: **Top-3** after re-sort

## Stage 3 — Fine Joint Refinement

Per beam candidate: scale  $\pm 0.3$  (step 0.3)  $\times$   $\theta \pm 0.6^\circ$  (step 0.3 $^\circ$ )

**NEW** **Multi-peak extraction:** Top-4 distinct spatial peaks per NCC surface (min\_sep 15% of template size) — prevents aliased periodic lobe from silently winning.

## Final Selection

All candidates scored by:

$$\text{confidence} = \text{score} \times (1 - \text{GAR})$$

Winner = highest confidence  $\rightarrow$  subpixel parabolic refinement

## Rejection Logic

$$\text{found} = (\text{score} \geq 0.43) \text{ OR } (\text{gar} \leq 0.65)$$

Score carries primary separation weight. GAR rescues borderline-score present pairs with high spatial confidence (low ambiguity).

Calibrated on real 20-pair organizer sample set. F1 = 1.000.

## Safety Budget

Wall-clock early-exit guard at **17.5s** prevents hard-timeout kills.

Median actual: **2.13s/pair**.

# The Single Biggest Fix: Core NCC Primitive

## ✗ Hand-Rolled FFT-NCC (Before)

Custom implementation using integral images + FFT correlation.

| METRIC         | VALUE                        |
|----------------|------------------------------|
| Location Error | 18 – 358 px                  |
| NCC Score      | 0.66 – 0.82                  |
| Status         | Wrong location at high score |

Verified directly on real organizer data at exact ground-truth (scale,  $\theta$ ). The algorithm was confident but *wrong* — the hardest class of bug to find.

## ✓ cv2.matchTemplate (After)

TM\_CCOEFF\_NORMED — OpenCV's verified NCC implementation.

| METRIC         | VALUE                   |
|----------------|-------------------------|
| Location Error | 0.3 – 0.7 px            |
| NCC Score      | 0.85 – 0.87             |
| Status         | Correct on all 20 pairs |

Same inputs, same (scale,  $\theta$ ). The problem was never search strategy, decoy ambiguity, or aliasing — it was the correlation primitive itself.

## Other Key Fixes

**SCALE STEP** 1.0 → 0.5 — true scale at 10.55 was 0.45 from nearest grid point, causing wrong-position lock

**ABSENT GENERATOR** Same-pitch decoy aliases → switched to mismatched-periodicity reference decoy (matching organizer design)

**THRESHOLD** Score gap of 0.0001 between p012/p018 → GAR provides 0.13 real margin for separation



# Verified on Real Organizer Sample Data

20/20

ALL PAIRS CORRECT

0.26 px

MEDIAN ERROR

F1 = 1.0

REJECTION SCORE

2.13s

MEDIAN TIME / PAIR

## Per-Set Breakdown

| SET         | DESCRIPTION        | ACCURACY | MEDIAN ERR             | MAX ERR |
|-------------|--------------------|----------|------------------------|---------|
| A (8 pairs) | Nominal pose       | 8/8      | 0.20 px                | 0.71 px |
| B (6 pairs) | Degraded           | 6/6      | 0.44 px                | 1.18 px |
| C (4 pairs) | Absent (rejection) | 4/4      | All correctly rejected |         |
| D (2 pairs) | RGB optical bonus  | 2/2      | 0.28 px                | 0.31 px |

## vs. Organizer Baseline (Naive ZNCC)

| METRIC         | OURS  | BASELINE |
|----------------|-------|----------|
| Set A accuracy | 1.000 | 1.000    |
| Set B accuracy | 1.000 | 0.467    |
| Set D accuracy | 1.000 | 1.000    |
| Rejection F1   | 1.000 | 0.897    |
| Median time    | 2.13s | —        |

Set B improvement is the strongest differentiator — degraded images are where beam search + cv2 NCC dramatically outperforms naive ZNCC.

# Self-Assessment Against Official Rubric

| COMPONENT                                           | MAX POINTS | ESTIMATED      | NOTES                                    |
|-----------------------------------------------------|------------|----------------|------------------------------------------|
| Localization<br>( $0.45 \times A + 0.55 \times B$ ) | 40         | ~39.6          | All within tier 1 except one tier 2      |
| Pose — Scale recovery                               | 10         | ~10            | Scale error <1% on all pairs             |
| Pose — Rotation recovery                            | 10         | ~10            | $\theta$ error <0.25° on all pairs       |
| Rejection F1                                        | 15         | 15             | F1 = 1.000 on sample set                 |
| Confidence calibration (AUC)                        | 10         | ~8–10          | Score separates outcomes cleanly         |
| Efficiency                                          | 5          | ~4–5           | 2.13s — well under 5s target             |
| Carried forward                                     | 10         | ~8–10          | Generator + citations + failure analysis |
| <b>TOTAL</b>                                        | <b>100</b> | <b>~94–100</b> |                                          |

## Localization Tiers (Official)

| ERROR (PX)  | CREDIT | OUR RESULTS       |
|-------------|--------|-------------------|
| $\leq 1$ px | 1.0    | 19/20 pairs       |
| $\leq 2$ px | 0.8    | 1/20 pairs (p010) |
| $\leq 3$ px | 0.6    | —                 |
| $\leq 5$ px | 0.4    | —                 |
| $> 5$ px    | 0.0    | —                 |

## Threshold Calibration

Critical boundary: **p012** (present, score=0.429) vs **p018** (absent, score=0.429)

Score gap: 0.0001 — but GAR provides **0.13 real margin**

p012 GAR=0.62 → found=1 ✓

p018 GAR=0.75 → found=0 ✓

# Failure Modes, Limitations & Explainability

## Failure Mode 1: Periodic-Array Ambiguity

Without prior window, structurally identical periodic sites produce NCC scores within 0.3% of each other. Image-content-only tie-break fails when two sites are genuinely, physically indistinguishable.

**Resolution:** Prior-window mechanism uses information *outside the image alone* (stage commanded position). **100% success with prior.**

## Failure Mode 3: FFT-NCC Implementation Bug

Hand-rolled integral-image + FFT correlation had a subtle bug — returned wrong location at 18–358 px error while reporting high confidence scores (0.66–0.82). The algorithm appeared to work but silently produced incorrect results.

**Resolution:** Replaced with `cv2.matchTemplate` — the single biggest accuracy fix. **0.3–0.7 px after fix.**

## Failure Mode 2: Scale Quantization

At `scale_step=1.0`, true scale 10.55 falls 0.45 from nearest grid point. Degraded NCC quality at wrong scale lets unrelated structural matches win outright — confident wrong positions at 81–226 px error.

**Resolution:** `scale_step=0.5` ensures  $\leq 0.25$  from true scale. **All sub-1px after fix.**

## Known Remaining Limitations

- ▶ Borderline score gap (0.0001) between hardest present/absent pair — may produce edge cases on full 200-pair test set
- ▶ RGB handled as grayscale conversion — no color-specific matching
- ▶ Efficiency depends on other teams for quartile ranking
- ▶ Local calibration data doesn't fully match organizer background diversity

# Code, Documentation & Reproducibility

## Phase 2 Codebase

- ▶ `register.py` — CLI entry point (organizer runs this)
- ▶ `localize_v2.py` — Beam-search NCC localizer
- ▶ `generate_dataset_v2.py` — Extended synthetic generator
- ▶ `scorer.py` — Self-assessment scorer
- ▶ `calibrate_*.py` / `sweep_thresholds.py` — Calibration tools

## Phase 1 Codebase (Preserved)

- ▶ `generate_dataset.py` — DRAM/FinFET/RGB generator
- ▶ `localize.py` — Classical FFT-NCC localizer
- ▶ `evaluate.py` — Batch evaluation harness
- ▶ `dl_localize.py` — Siamese CNN ablation

## Documentation

- ▶ `docs/proposal.pdf` — Full technical proposal
- ▶ `docs/citations.md` — 15+ annotated sources
- ▶ `README.md` — Setup + run instructions

## Environment

- ▶ Python 3.12, NumPy 2.5, SciPy 1.18, OpenCV 5.0
- ▶ No GPU, no deep learning at inference
- ▶ No network access required
- ▶ 4-core CPU, 8 GB RAM

## GitHub

**[github.com/YashKothari06/Drift-sense](https://github.com/YashKothari06/Drift-sense)**

All Phase 1 + Phase 2 code, docs, sample outputs, and trained weights.

PHASE 2 COMPLETE

# Sub-Pixel Accuracy. Real-Time Speed. Industry-Grounded.

**100%**

SAMPLE ACCURACY

**0.26 px**

MEDIAN ERROR

**F1 = 1.0**

REJECTION

**2.13s**

PER PAIR

**15+**

CITATIONS

Classical FFT-NCC + multi-stage beam search + cv2.matchTemplate  
beating the organizer baseline on *all* scoring axes.

**Team Blitzkrieg**

Bhushan Thatte & Yash Kothari · BITS Pilani

Thank You